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A Review of Research Progress on Durian Maturity Identification Technology

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***Abstract*—The maturity identification of durian is crucial for ensuring the quality and market value of the fruit. It is an important tropical economic fruit. The research progress of durian maturity identification technology has been systematically summarized. Firstly, traditional detection methods were introduced, including sensory evaluation method, physicochemical index detection method and biochemical index analysis. The advantages and disadvantages of these methods were analyzed. Then modern intelligent detection technologies were elaborated in detail, including computer vision technology, acoustic detection technology, spectral analysis technology, chemical marker detection and molecular biology technology. The principles, application methods and performance characteristics of these technologies were analyzed with emphasis. Finally, the development trend of multi-source information fusion, especially the fusion technology of computer vision and acoustic detection, as well as the existing problems and future prospects were discussed. Research shows that the fusion technology based on computer vision and acoustic detection is becoming the main development direction for durian maturity identification. This fusion technology integrates appearance features and internal texture features, and can provide a more comprehensive and accurate maturity assessment solution for the modernization development of the durian industry, as well as technical support. It provides important technical support for the development of durian maturity assessment.**

Keywords—Durian Maturity, Non-destructive testing, Multi-source information fusion, Deep learning

I. INTRODUCTION

Durian is an important tropical fruit, renowned for its unique aroma and taste [1]. Durian is harvested before the start of the climatic maturity period, but the physiological maturity period varies by variety within 120 to 130 days after the fruit harvest. This information is necessary for achieving the best fruit quality, as unripe fruits cannot ripen independently [4]. During the growth and ripening process of the "Monthong" durian (Durio Zibethinus Murr) fruit, the dry weight of the pulp, soluble solids, total sugar and reducing sugar contents reached the maximum level at 127 days (DAA) after pollen ripening [18]. Accurately judging the ripeness of durian is of great significance for ensuring fruit quality, extending shelf life and meeting consumers' demands [2]. The traditional maturity judgment methods mainly rely on human experience, which has disadvantages such as strong subjectivity and low efficiency. In recent years, with the rapid development of technologies such as computer vision and acoustic detection, significant progress has been made in non-destructive testing of durian maturity. Especially the integration of computer vision and acoustic detection technology can not only evaluate the appearance characteristics of durian, but also provide internal quality information, offering a new idea for the comprehensive assessment of durian maturity [3]. This paper will review the research progress of durian maturity identification technology from two aspects: traditional detection methods and modern intelligent detection technologies, and focus on discussing the development trend and application prospects of the fusion technology of computer vision and acoustic detection.

II. TRADITIONAL DETECTION METHODS

A. Sensory Evaluation Method

Sensory evaluation method is a traditional way to determine the ripeness of durian, mainly relying on human sensory experience [2]. This method includes observation of appearance features, judgment of knocking sounds, odor assessment, observation of the shed layer and SAP observation, etc.

The observation of appearance features mainly includes indicators such as dazzling color, shape and arrangement, color and state of the fruit stalk, depth and color of the fruit groove, and the number of flowering days (DAA). The DAA range varies among different varieties of durian and is an important indicator of maturity.

Knocking sound judgment is a method of judging the maturity of fruits by knocking and listening to the sound [4]. Modern acoustic detection technology has further systematized this method, improving the objectivity and accuracy of the judgment.

Odor assessment is based on the principle that durians at different maturity stages have different aroma characteristics [5]. Different varieties of durians also have differences in aroma characteristics, providing a basis for variety identification.

Exfoliation layer observation is a method to determine the degree of maturity by observing the characteristics of the exfoliation layer at the junction of the fruit stalk and the fruit. SAP observation is to assess the degree of maturity by observing the characteristics of the SAP at the incision of the fruit stalk.

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Although sensory evaluation methods are simple and feasible, they have limitations such as strong subjectivity, low efficiency, and reliance on experience. Preliminary experimental results show that when distinguishing some mature stages, the accuracy of these traditional methods is relatively low.

B. Physical and Chemical Index Detection Method

The physical and chemical index detection method is an objective approach for assessing the maturity of durian, mainly including dry matter content, soluble solid content, hardness, pH value, organic acids, and aroma components, etc. [6]

Dry matter is the accumulation of soluble and insoluble carbohydrate content [19,20]. Dry matter content is an important indicator of durian maturity, which is related to the soluble sugar content and starch accumulation in the fruit and can predict the future edible quality of the fruit. The drying method is usually adopted for determination. The durian flesh sample is dried at 105℃ to a constant weight, and the percentage of dry matter content is calculated. Studies show that the dry matter content is positively correlated with the ripeness of durian, and mature durian usually has a higher dry matter content. The dry matter content of the Monthong variety of durian usually reaches its optimal harvest period 100 to 110 days after flowering, when the dry matter content is approximately 30 to 32%.

The content of soluble solids is usually measured by a refractometer in the juice of durian flesh, expressed as ° Brix, which reflects the sweetness of durian and is closely related to its maturity. Ripe durian usually has a very high content of soluble solids. As durian ripens, starch gradually hydrolyzes into sugar, leading to an increase in soluble solid content. The soluble solid content of Mengdong durian can reach 30-35 ° Brix 100 days after flowering.

The hardness of durian flesh is usually determined by a texture analyzer, measured in Newtons (N) [8]. The hardness is negatively correlated with the maturity of durian. As the maturity gradually decreases, this change in hardness is related to the enhanced activity of cell wall degrading enzymes, resulting in a loose cell wall structure and a softer texture.

The pH value of durian pulp juice is usually measured by a pH meter. The pH value has a certain correlation with the maturity of durian, but the pH value variation range of different durian varieties may vary. The main organic acids in durian include citric acid, malic acid, oxalic acid, etc. As the maturity increases, the content of organic acids gradually decreases, and the pH value also increases accordingly[9]. The unique aroma of durian comes from volatile sulfides, such as disulfide and allyl sulfide, etc. The aroma components are usually analyzed by gas chromatography-mass spectrometry (GC-MS). As durian matures, the content of characteristic aroma substances gradually increases and the aroma intensity intensifies. The aroma component composition of durian varies significantly among different varieties and can be used as an indicator to identify the variety.

Although these methods are objective, they belong to destructive testing, are complex to operate, and are not suitable for rapid detection. Moreover, these methods require professional instruments and equipment as well as operational skills, and are not suitable for rapid application in the field or packaging factories.

C. Biochemical Index Analysis

During the ripening process of durian, a series of biochemical changes occur, which can serve as an important basis for judging maturity [10], mainly including changes in enzyme activity, plant hormone levels, and gene expression regulation, etc.

As durian ripens, the activity of polyculatide acid e (PME) first increases and then decreases, catalyzing the demethylation of pectin and affecting the softening degree of the fruit. The gradually increasing activity of polygaluronidase (pg) during ripening catalyzes the degradation of pectin chains, resulting in a loose cell wall structure. Meanwhile, with the increase of maturity, the activities of α -amylase and β -amylase are enhanced, promoting the hydrolysis of starch into sugar, and the sweetness of the fruit also increases accordingly.

Durian is a fruit with vigorous respiration. When it ripens, the release of ethylene first increases and then decreases. During ripening, the content of abscisic acid (ABA) gradually increases, which softens the fruit and promotes the synthesis of aroma substances. However, as the ripening increases, the content of auxin (IAA) gradually decreases, thereby relieving the inhibition of ethylene synthesis.

When durian ripens, the expression of genes related to ethylene synthesis, such as ACC synthase (ACS) and ACC oxidase (ACO), is increased. At the same time, the expression levels of cell wall degrading enzyme genes such as PG, PME, and cellulase are also increased, promoting fruit softening. The genes related to sugar metabolism also changed. The expression of the starch synthase gene was down-regulated, and the expression of the starch decomase gene was up-regulated. Studies have shown that the DOF transcription factor family is closely related to the ripening process of durian fruits and participates in regulating the development and ripening of fruits.

The above biochemical indicators can reflect the maturity of durian at the molecular level, but the determination method is rather complex, requiring professional instruments and equipment as well as advanced technology, which is not conducive to promotion and application.

III. Modern Intelligent Detection Technology

A. Computer Vision Technology

The steps of applying computer vision technology in durian maturity identification include image acquisition, image preprocessing, feature extraction and classification recognition, etc. [12]

During the picture shooting stage, it is necessary to control the lighting conditions and shooting angles, and at the same time, use a digital camera or industrial camera to take pictures of durian. Only in this way can the picture quality be guaranteed. Image preprocessing includes color space conversion (such as RGB, HSV, Lab, etc.), image enhancement

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(such as histogram equalization, filtering, etc.), and image segmentation (extraction of durian fruit regions). In the feature extraction stage, color features (such as color moments, color histograms, etc.), texture features (such as GLCM, LBP, etc.) and shape features (such as area, perimeter, roundness, etc.) are extracted from the processed image. Finally, in the classification and recognition stage, traditional machine learning algorithms (such as SVM, KNN, etc.) or deep learning algorithms (such as CNN, ResNet, etc.) are used to classify the extracted features to identify the maturity of durians, with an accuracy rate of over 90%.

The advantages of computer vision technology lie in its non-destructive nature, rapidity, and high degree of automation. It can effectively analyze the appearance change characteristics of durians, such as dazzling colors, arrangements, and shapes. In addition, through the application of deep learning algorithms, computer vision technology can automatically learn and extract the deep-level features related to the maturity of durian, reducing the reliance on manual feature engineering. However, this method is greatly affected by lighting conditions and mainly relies on appearance features for recognition, thus having limited ability to evaluate the internal quality.

B. Spectral Analysis Technology

The application of spectral analysis technology in the identification of durian maturity mainly includes near-infrared spectroscopy technology, hyperspectral imaging technology and Raman spectroscopy technology, etc. Specifically, there are the following aspects: [13, 17].

Near-infrared spectroscopy technology uses the spectral information generated by the interaction between near-infrared light and sample molecules to evaluate the quality of durian. It mainly employs a Fourier transform near-infrared spectrometer for measurement. The application methods include directly scanning the durian flesh or scanning the entire durian fruit. Data analysis usually employs algorithms such as Partial least squares regression (PLSR). Studies have shown that by scanning the flesh of durian, the dry matter content can be predicted, with a verification coefficient of determination (R) of up to 0.89 and a predicted mean square root error (RMSEP) of 3.60%. By scanning the complete fruits, the content of soluble solids in the pulp can be predicted. Its verification coefficient of determination (r) can reach 0.82, and the predicted mean root error (rmseph) is 1.23° brix, that is, the core mass index (RMSEP).

The hyperspectral imaging (HSI) system combines spectral and image analysis techniques to simultaneously obtain spatial and spectral information of durian. It mainly uses a push-sweep near-infrared hyperspectral imaging system (900-1600 nanometers) for measurement. The application methods include scanning the durian flesh to obtain the spectral data of each pixel point, as well as analyzing the spatial distribution characteristics of durian flesh at different maturity levels, etc. Data analysis typically employs classification algorithms such as Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM), as well as regression algorithms like Partial Least Squares Regression (PLSR). Studies show that by using the LDA calculation method, the ripening stages (unripe, ripe and over-ripe) of durian flesh can be classified, and the accuracy rate can reach 100%. The dry matter content of pulp can be predicted by using PLSR calculation. The predicted coefficient of determination (Rp) is greater than 0.80, and the predicted mean root error (RMSEP) is less than 1.6%.

Raman spectroscopy technology, based on molecular vibrational and rotational energy level transitions, can provide molecular structure information of samples [15]. This technology can offer information on the molecular structure of durian, providing new ideas for maturity determination. Raman spectroscopy technology is non-destructive, rapid and does not require sample pretreatment, making it suitable for durian maturity assessment.

The advantage of spectral analysis technology lies in its ability to rapidly and non-destructively assess the internal quality of durian, such as dry matter content and soluble solid content. However, this method requires professional instruments and equipment as well as operational skills, and is affected by factors such as sample state and environmental conditions.

C. Acoustic Detection Technology

Acoustic detection technology is a non-destructive method that can assess the internal quality of durian [16]. This technology includes links such as acoustic signal acquisition, feature parameter extraction and pattern recognition analysis.

Acoustic signal collection mainly uses a specially designed hammer or fingers to tap the durian fruit to generate acoustic signals. Signal acquisition devices usually use microphones or acceleration sensors. The sampling frequency is usually 44.1kHz or higher to ensure signal quality. In the stage of feature parameter extraction, time-domain features (such as peak, mean, variance, kurtosis, etc.), frequency-domain features (such as spectral energy, principal frequency, frequency band energy ratio, etc.) and time-frequency features (such as wavelet transform coefficients, short-time Fourier transform coefficients, etc.) are extracted from the collected acoustic signals. Pattern recognition analysis uses classification algorithms such as artificial neural networks (ANN), support vector machines (SVM), K-nearest neighbors (KNN), etc. to classify the extracted features and identify the maturity of durians. Studies show that by using acoustic features combined with the ANN algorithm, the maturity of durian can be classified, and the accuracy rate can reach more than 80%.

D. Chemical Labeling Detection

Chemical marker detection is an evaluation method based on the maturity of chemical components in the stem and flesh of durian [5]. The technology includes the extraction of chemical markers, the analysis of key chemical markers, and the establishment of predictive models, etc.

The extraction of chemical markers involves collecting the stems or flesh of durians, and conducting freeze-drying and grinding treatments. The extraction method usually uses organic solvents such as methanol and ethanol to extract the chemical components in the sample. Analytical instruments include gas chromatography-mass spectrometry (GC-MS), high performance liquid chromatography (HPLC), nuclear magnetic resonance spectroscopy (NMR), etc.

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The key chemical markers in the stem of durian include sucrose (which is positively correlated with the maturity of durian and is one of the important chemical markers, with a content of 2-3 mg/g dry weight in the mature stem of durian) and asparagine (which is the most abundant and significantly positively correlated with the maturity of durian). The content of arginine (which is positively correlated with the maturity of durian and is a potential maturity indicator, with the content increasing by more than 50% with the increase of maturity) and piperidine acid (which is positively correlated with the maturity of durian and can be used as an auxiliary indicator for maturity prediction) in the fruit stem of mature durian can reach 4-6 mg/g dry weight.

The characteristic components in durian flesh include sugar components (the sucrose content in mature durian flesh can reach 15-20% (fresh weight), accounting for 65-80% of the total sugar, followed by fructose and glucose), and organic acids (the main organic acids include citric acid, malic acid and oxalic acid). The total acid content decreases with the increase of maturity), fatty acids (durian flesh is rich in unsaturated fatty acids such as oleic acid and linoleic acid, accounting for 60-70% of the total fatty acids), and sulfides (the characteristic aroma of durian mainly comes from sulfur-containing compounds, such as diallyl disulfide and diallyl trisulfide, etc.).

The establishment of the prediction model includes correlation analysis (analyzing the correlation between chemical markers and durian maturity), principal component analysis (using the PCA method to conduct dimensionality reduction analysis on multiple chemical components and identify key components), and prediction models (using methods such as multiple regression analysis and partial least squares regression (PLSR) to establish prediction models between chemical markers and maturity). Studies show that chemical markers can be used to predict the ripeness of durian, with a prediction accuracy rate of over 85%.

The advantage of chemical marker detection lies in its ability to provide objective maturity indicators and is closely related to the physiological state of durian. However, this method requires professional instruments and equipment as well as operational skills, and it is a destructive test.

E. Molecular Biology Techniques

The application of molecular biology techniques in durian variety identification and maturity assessment mainly includes DNA probe technology, gene expression analysis and transcription factor research, etc. [11]. This technique encompasses multiple links such as DNA probe technology, gene expression analysis and transcription factor research.

DNA probe technology utilizes the complementary pairing of specific DNA sequences with target genes to identify durian varieties and genes related to ripening. The application methods include PCR amplification (amplifying the target DNA sequence using specific primers) and electrochemical biosensors (fixing the amplified DNA sequence on the electrode surface and detecting the target sequence through electrochemical signals). Studies have shown that the use of electrochemical biosensors can rapidly identify the Musang King Durian variety, with a detection limit of up to 0.1 N / μ L, high sensitivity and specificity.

Gene expression analysis includes transcriptome sequencing (analyzing the gene expression profiles of durian at different maturation stages), differentially expressed genes (identifying differentially expressed genes related to the maturation process), and functional annotation (performing functional annotation and pathway analysis on differentially expressed genes). The research on transcription factors mainly focuses on the Dof gene family (studies have found that the Dof (DNA binding with one finger) transcription factor family is closely related to the ripening process of durian fruits). Through genomic analysis, 24 Dof genes were identified. Analyze the expression patterns of the Dof gene in different varieties and at different maturity stages. Through functional verification, the role of the Dof gene in the ripening process of durian fruits was confirmed.

The advantage of molecular biology technology is that it can understand the ripening process of durian at the molecular level and provide new ideas for assessing the degree of ripening. However, this method requires professional instruments and equipment as well as operational skills, is complex to operate, and is not suitable for rapid detection.

IV. Technological Development Trends and Prospects

A. Computer vision and Acoustic Detection Integrating Technology

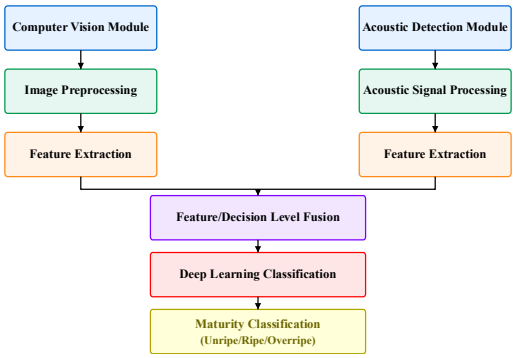


Fig. 1. Integration framework of computer vision and acoustic detection for durian maturity identification.

The fusion technology of computer vision and acoustic detection is an important branch of multi-source information fusion. By combining the advantages of computer vision and acoustic detection technology, the appearance characteristics and internal quality information of durian are obtained simultaneously to improve the accuracy and reliability of maturity judgment[3].As shown in Figure 1, the integration framework combines features from both computer vision and acoustic detection to achieve superior maturity identification.

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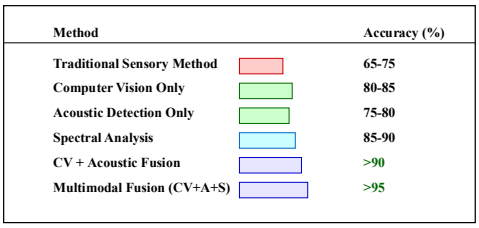


Fig. 2. Comparison of maturity identification accuracy for different methods.

Figure 2 demonstrates that the fusion approaches achieve significantly higher accuracy than individual methods, with CV + Acoustic Fusion reaching above 90% accuracy, and multimodal fusion exceeding 95%.

1) Fusion methods and strategies

The fusion technology of computer vision and acoustic detection mainly adopts two strategies: feature-level fusion and decision-level fusion. Feature-level fusion is to combine the feature vectors extracted from images and acoustic signals to form a joint feature vector containing multimodal information, and then use machine learning algorithms for classification and recognition. Decision-level fusion involves using computer vision and acoustic detection technologies respectively to determine maturity, and then integrating multiple decision results through methods such as voting, weighting, or probabilistic fusion.

2) Application of deep learning in fusion technology

Deep learning technology provides new concepts and approaches for the integration of computer vision and acoustic detection. Convolutional Neural networks (CNN) and recurrent neural networks (RNN) are respectively applicable to the processing of images and time series signals. Through the design of multimodal deep learning networks, the joint learning and feature extraction of visual and acoustic information can be achieved. The introduction of the attention mechanism enables the model to adaptively focus on the important features of different modalities, further improving the fusion effect. Studies show that in the task of durian maturity classification, the fusion technology of computer vision and acoustic detection based on deep learning performs excellently, and its accuracy can reach more than 90%. And it can achieve adaptation and integration of the important features of different modes in different modes.

3) Portable fusion detection equipment

The development of portable fusion detection devices has made the fusion technology of computer vision and acoustic detection more suitable for use in fields and packaging plants [7]. These devices usually integrate some components, such as micro cameras, microphones or acceleration sensors, embedded processing systems, and wireless communication modules, etc. Users can obtain real-time evaluation results of durian ripenness through smartphone apps or cloud platforms. These portable devices are easy to operate, have moderate costs, and have promising application prospects. Users can conduct real-time evaluations through mobile apps or cloud platforms.

The advantage of the fusion technology of computer vision and acoustic detection lies in that it can conduct a comprehensive assessment of the appearance characteristics and internal quality of durian, and provide a more comprehensive and accurate judgment in terms of maturity. This technology inherits the core ideas of visual observation and knocking sound judgment in traditional sensory evaluation methods, but achieves automation and standardization through modern information technology, and has higher objectivity and consistency.

B. Other Multi-Source Information Fusion Technologies

In addition to the fusion of computer vision and acoustic detection, other multi-source information fusion technologies have also been applied in the identification of durian maturity.

Visual-spectral fusion combines computer vision and spectral analysis techniques to simultaneously obtain information on the appearance and internal quality of durians. The fusion strategies include feature-level fusion, decision-level fusion, etc. Studies show that visual-spectral fusion can improve the accuracy rate of durian maturity classification, which can reach more than 95%.

Multimodal data fusion combines multiple detection techniques, such as computer vision, spectral analysis, acoustic detection, and chemical marker detection, etc. The fusion strategies include feature-level fusion, decision-level fusion, deep learning fusion, etc. Multimodal data fusion can comprehensively utilize the advantages of different detection methods to improve the accuracy and reliability of maturity judgment.

V. Conclusion

Durian maturity identification technology is developing towards intelligence, automation and portability. Meanwhile, the fusion technology of computer vision and acoustic detection, as a comprehensive method capable of simultaneously evaluating the appearance characteristics and internal quality of durians, has shown broad application prospects. This fusion technology inherits the advantages of visual observation and knocking sound judgment in the traditional sensory evaluation method. Through modern information technology, it has achieved automation and standardization, improving the accuracy and consistency of maturity judgment.

With the advancement of deep learning and multimodal fusion algorithms, the performance of the fusion technology of computer vision and acoustic detection has been significantly improved, providing strong support for the standardized and modernized development of the durian industry. Especially the fusion model based on deep learning shows excellent adaptability in dealing with complex environments and diverse varieties, with an accuracy rate of over 90%.

The development of portable integrated detection equipment has made this technology closer to the practical application requirements, providing convenient technical tools for on-site management and harvest decision-making. The integration technology of computer vision and acoustic detection, through the combination with Internet of Things, cloud computing and big data technologies, is expected to

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become an important component of smart agriculture and also an important part of food supply chain management.

In the future, the fusion technology of computer vision and acoustic detection will further improve the accuracy, efficiency and reliability in maturity recognition with the development of algorithm optimization, multimodal deep learning, adaptive fusion mechanisms and standardized systems. Taking into account the technical performance, cost and practicality comprehensively, this integrated technology is expected to become the mainstream method for identifying the maturity of durians, providing strong technical support for the high-quality development of the durian industry.

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